

AUTOMATIC EXTRACTIVE AND GENERIC DOCUMENT SUMMARIZATION BASED ON NMF

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Submitted: 9th March 2022; Accepted: 19th October 2022

Abstract

Nowadays, textual information grows exponentially on the Internet. Text summarization (TS) plays a crucial role in the massive amount of textual content. Manual TS is time-consuming and impractical in some applications with a huge amount of textual information. Automatic text summarization (ATS) is an essential technology to overcome mentioned challenges. Non-negative matrix factorization (NMF) is a useful tool for extracting semantic contents from textual data. Existing NMF approaches only focus on how factorized matrices should be modeled, and neglect the relationships among sentences. These relationships provide better factorization for TS. This paper suggests a novel non-negative matrix factorization for text summarization (NMFTS). The proposed ATS model puts regularizes on pairwise sentences vectors. A new cost function based on the Frobenius norm is designed, and an algorithm is developed to minimize this function by proposing iterative updating rules. The proposed NMFTS extracts semantic content by reducing the size of documents and mapping the same sentences closely together in the latent topic space. Compared with the basic NMF, the convergence time of the proposed method does not grow. The convergence proof of the NMFTS and empirical results on the benchmark data sets show that the suggested updating rules converge fast and achieve superior results compared to other methods.

Keywords: Text Summarization, Latent Topics, Non-negative Matrix Factorization, Updating Rules

1 Introduction

The Internet resources such as blogs, user reviews, websites, and social media networks are massive textual data. Also, there is a tremendous of textual content on the other archives of news books, novels, articles, etc. The textual information on the Internet and different archives increase exponentially. So, users need a lot of time to find the content in the searching process. There are many redundant amounts of the resulting texts [1]. Thus, summariz-

ing the document resources is a crucial task. Manually summarizing is time-consuming and an expensive task [2]. Automatic text summarization (ATS) plays an important role to deal with the mentioned challenges.

The purpose of an ATS system is to create a summary that briefly contains the main concepts in the original document [3]. The ATS systems help users to generate summaries automatically to save a lot of time. The created summary contains the

most meaningful information in the input text and should be shorter in length than the original document. Text summarization (TS) approaches can be grouped into query-based summarization and generic summarization. Query-based summarization generates the summaries that are closely related to the query while generic summarization creates the summarized document and offers the significant content of the input texts [4].

ATS systems are proposed based on one of the extractive, abstractive, or hybrid methods. The extractive techniques choose the most important sentences from the given document and utilize them to create the summary. The abstractive methods represent the original document in an intermediate form then produce the summary by paraphrasing sentences that differ from the given text sentences. The hybrid techniques incorporate both the extractive and abstractive methods. Extractive approaches are more popular [5, 6] than abstractive techniques [7] since the latter are more complex in terms of language understanding and demand natural language generation.

ATS approaches can be grouped into two categories: supervised and unsupervised techniques. Supervised approaches use large volumes of man-made summaries for training, and are proper for texts that are relevant to the model. Therefore, they can not create a useful summary for texts that are not similar to the train data. Also, these methods require to be re-train when users change the purpose of summarization or the characteristics of texts. Unsupervised techniques do not need training data and apply predetermined criteria to score sentences and use high-scoring sentences in summary. These approaches include optimization, graph-based, statistical, probabilistic, machine learning, and matrix factorization (MF) [6].

Optimization methods solve the TS problem by modeling it as an objective function based on summary features such as relevance, length, and redundancy [6]. However, these techniques are computationally expensive due to the cost of simultaneously optimizing the parameters. Graph-based representation is another technique for TS [5, 8]. These approaches transform a document into a graph, where each node is a sentence and semantically related sentences are connected with edges. Then, node-based ranking algorithms are

used for choosing summary sentences. Language and domain-independent attributes of these methods make them attractive. Statistical methods utilize features such as frequency of important words, cue words, sentence position, sentence length, etc as attributes to score and choose sentences for including in the summary [9]. These methods are simple and computationally efficient but neglect the semantic of content. Probabilistic techniques for TS include conditional random fields and hidden Markov models, which assume sentence interactions and dependencies to find significant sentences [10]. These approaches extract contextual information from the text but have high computational complexity. Newly, neural network (NN) and reinforcement learning-based methods attract more attention [11]. Although NN techniques create high-quality summaries, these approaches require large datasets for training, extensive computation, and suffer from low interpretability. These techniques often depend on a specific collection and domain therefore less acceptable for generic TS. Matrix factorization methods like principal component analysis (PCA), latent semantic analysis (LSA), and non-negative matrix factorization (NMF) generate promising results for TS [12, 13]. These techniques are popular because of cheap resource requirements. Additionally, matrix factorization approaches for TS are independent of language, which makes them interesting for generic TS.

This paper proposes a novel generic, extractive, and unsupervised text summarization method based on NMF. The proposed approach explicitly applies the pairwise relationships among sentences as penalties on factorized matrices to learn the semantic content of texts. This paper presents a new cost function based on the Frobenius norm and incorporates the pairwise relationships into it. This paper also develops an optimization scheme to minimize the cost function by proposing novel iterative update rules. This leads to a new parts-based representation that respects the relationships among sentences.

- This paper presented a novel summarization method based on NMF to learn the semantics of documents.
- The proposed approach uses relationships among sentences to better factorization. The

proposed method preserves the expected pairwise relations between sentences remaining after decomposition.

- Frobenius norm is applied as a penalty by considering the algorithm's convergence through numerous experiments.
- Complete and sufficient proof is supplied for the proposed updating rules of the cost function.
- The time complexity analysis showed that the proposed method does not raise convergence time compared with the basic NMF.
- Experiments show that NMFTS achieves superior performance compared to other methods.

The remainder of the paper is organized as follows: Section 2 presents the background of and related works. The proposed method and convergence proof are explained in Section 3. Section 4 discusses experiments. The conclusion is given in Section 5.

2 Background and related works

2.1 ATS categorization

TS approaches are grouped based on the approach, summary, and summarization algorithm. TS is classified based on the approach to abstractive, extractive, or hybrid. The abstractive TS technique shows the texts in an intermediate form, and the summary is created from this intermediate form. The extractive TS method chooses the most significant sentences in the given texts, and the summary is formed by connecting the chosen sentences. Different from extractive methods, abstractive summaries include sentences that are different from the given text. The hybrid TS method combines both extractive and abstractive techniques [14].

Text summarization is grouped based on the nature of the summary to generic or query-based. Generic methods extract essential information from texts to create a general sense of its context [4, 15]. Query-based TS aims to create a summary of a document that directly answers or is pertinent to the query, while generic methods provide an overall sense for the text context [15, 16].

TS methods are classified based on the algorithm to supervised and unsupervised. The supervised methods require a training dataset. Human efforts are needed to manually provide the training dataset, and it is time-consuming. On the other hand, the unsupervised approaches do not need a training phase and training dataset [17].

2.2 Non-negative matrix factorization

NMF is reported in the literature as an optimization method to learn the parts-based presentation of objects [18, 19, 20]. In this method, the elements of the factorized matrices must be non-negative. Consider document D includes N sentences and M terms after removing stop-words; NMF factorizes a sentence-term matrix $D_{N \times M}$ to learn two non-negative sentences matrix $S_{N \times K}$ and terms matrix $T_{K \times M}$ ($K < \min\{N, M\}$) where the product is a fine approximation of the texts matrix, $D \approx ST$. KL -divergence and Frobenius norm are commonly employed cost functions to measure the approximation. Frobenius norm computes the squared Euclidean distance between two matrices as follows:

$$\mathcal{F} = \sum_{n=1}^N \sum_{m=1}^M (D_{nm} - \sum_{f=1}^K S_{nf} T_{fm})^2 \quad (1)$$

KL -divergence is the asymmetric divergence:

$$\mathcal{KL} = \sum_{n=1}^N \sum_{m=1}^M (D_{nm} \log \frac{D_{nm}}{(ST)_{nm}} - D_{nm} + (ST)_{nm}) \quad (2)$$

These cost functions are convex regarding either S or T , but not both. Hence it is impossible to find the global minimum. Two methods are suggested to solve the cost functions of $\mathcal{F}$ and $\mathcal{KL}$ [?]. The following iterative update rules minimize the cost function $\mathcal{F}$:

$$S_{nf} = \frac{S_{nf}(DT^T)_{nf}}{(STT^T)_{nf}} \quad (3)$$

$$T_{fm} = \frac{T_{fm}(S^T D)_{fm}}{(S^T ST)_{fm}} \quad (4)$$

The updating rules to minimize $\mathcal{KL}$ are as follows:

$$S_{nf} = \frac{S_{nf} \sum_{m=1}^M T_{fm} D_{nm} / (ST)_{nm}}{\sum_{m=1}^M T_{fm}} \quad (5)$$

$$T_{fm} = \frac{T_{fm} \sum_{n=1}^N S_{nf} D_{nm} / (ST)_{nm}}{\sum_{n=1}^N S_{nf}} \quad (6)$$

The convergence proof of these iterative update rules is presented by Lee and Seung for equations (1, 2) [21]. NMF estimates the m -th column of D by a linear combination of the rows of S weighted by the K entries of column T_m , $D_m \approx ST_m$. Matrix S can be considered to be the basis for optimizing the linear compound of the sentences. Since a few basis vectors are applied to represent all vectors, an adequate estimation is acquired if the basis vectors learn the semantic content of the textual data. In TS, determining the latent space dimension is challengeable, and NMF transforms sentences to K -dimensional semantic topics. Therefore, a high-dimensional sentence-term matrix is represented with low-dimensional vectors. Unlike other MF methods, non-negative constraints only permit additive, not subtractive, combinations which is the reason for considering NMF as the parts-based representation [18].

The NMF has been applied in several applications. In [22], the authors suggested graph regularized NMF (GNMF) to image clustering by generating a similarity matrix, where the data points in various clusters are not similar. The distribution of data points is not in this way in TS. GNMF utilizes the similarities among decomposed vectors for dimension reduction. This technique attempts to keep similarities between images in the given space and the latent space. A semi-supervised MF is suggested to factorize partially labeled data [23]. The authors assumed data points have the same representation if they are in the same group. This method employs the relation between decomposed vectors and represents the relationships in the structure of factorized matrices. The authors used partial labels as constraints to image clustering and applied a semi-supervised method in the non-negative factorizing process. This setting has limitations in TS. A regularized single-element NMF method is presented to create a low-rank approximation of the given data [24]. This technique assumes the non-negativity constraints to decrease dimensions and

time complexity. The authors suggested a method called ANLF to create a non-negative latent factor [25]. This method learns the latent factor by single-element-dependent optimization.

MF methods have the potential to reveal the latent semantic structure of documents in the low dimensional space. MF approaches such as LSA, PCA, and NMF have demonstrated promising results for TS [13, 26, 27], topic modeling [28], and text clustering [19]. The authors used LSA for single TS by decomposing the sentence-term matrix and ranking sentences for inclusion in summary [27]. LSA factorizes the given data into factors that include both negative and positive elements. In [13], the authors successfully utilized NMF for ATS. Since the achieved factors by NMF are non-negative, the relations of the terms and topics in the text potentially improve understanding of the latent semantic space [18].

The research motivation of this paper was to create the theoretical analysis to understand how the proposed NMFTS summarizes documents; and from this perception, evaluate its performance on TS. The suggested approach uses the pairwise similarity between sentences as a regularizer for creating a cost function. The main idea is that the sentences with the same concept should be near to each other in the latent semantic space. This paper presents novel iterative update rules as an optimization method to minimize the cost function to discover the semantic topics of sentences. This paper has applied the Frobenius norm as a cost function, and the Kullback-Leibler is imposed as a constraint. The suggested update rules utilize relations between sentences to conduct a better decomposition. This results in a new parts-based representation regarding the relations among sentences. This paper provides convergence proof for the suggested update rules. The analysis of computation complexity has revealed that NMFTS does not raise convergence time by applying constraints. Also, NMFTS improves the space complexity by decomposing the sentence-term matrix into two low-dimension matrices.

3 Method

This paper suggests a novel algorithm, NMFTS, to learn the semantic structure of sentences.

NMFTS utilizes the similarity between sentences as regularizers. The similarity among sentences forms a similarity matrix. Cosine similarity is a common metric between two data points [29]. It can be computed in a multi-dimensional space. Cosine similarity is widely used in text mining [30]. The similarity weight for the two sentences is defined as follows:

$$\mathbb{S}(n, n') = D_n \cdot D_{n'} / (\|D_n\| \cdot \|D_{n'}\|) \quad (7)$$

where $\mathbb{S}(n, n')$ is the similarity weight between sentences n and n' . D_n and $D_{n'}$ are the M -dimensional term-frequency vectors. The KL -divergence can be used to measure the dissimilarity between term vectors of sentences. Assuming the sentences' similarity matrix, the following equation can be applied to calculate dissimilarity among sentences vectors.

$$\mathcal{R} = \sum_{n=1}^N \sum_{n'=1}^N \sum_{f=1}^K (S_{nf} \log \frac{S_{nf}}{S_{n'f}} + S_{n'f} \log \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') \quad (8)$$

If two sentences n and n' are close, then S_n and $S_{n'}$ should be similar to each other by minimizing Eq. (8). Adding this regularizer to the Frobenius norm outcomes the proposed objective function. Given a non-negative sentence-term matrix, NMFTS attempts to learn two non-negative matrices by optimizing the following objective function.

$$\begin{aligned} O = & \sum_{n=1}^N \sum_{m=1}^M (D_{nm} - \sum_{f=1}^K S_{nf} T_{fm})^2 + \\ & \lambda \sum_{n=1}^N \sum_{n'=1}^N \sum_{f=1}^K (S_{nf} \log \frac{S_{nf}}{S_{n'f}} + S_{n'f} \log \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') \end{aligned} \quad (9)$$

Subject to : $\lambda, S, T \geq 0$

The λ controls the smoothness of the new representation. The objective function O is convex concerning S or T , but not both. So, it is impossible to find a global minimum for this objective function. This paper presents an iterative algorithm that can find a local minimum.

The update rules for S and T are needed to minimize the objective function in Eq. (9). The update rules are created and solved by the related Lagrange function:

$$\begin{aligned} \mathcal{L} = & \sum_{n=1}^N \sum_{m=1}^M (D_{nm} - \sum_{f=1}^K S_{nf} T_{fm})^2 \\ & + \lambda \sum_{n=1}^N \sum_{n'=1}^N \sum_{f=1}^K (S_{nf} \log \frac{S_{nf}}{S_{n'f}} + S_{n'f} \log \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') \\ & + \sum_{n=1}^N \sum_{f=1}^K \alpha_{nf} S_{nf} + \sum_{m=1}^M \sum_{f=1}^K \beta_{fm} T_{fm} \end{aligned} \quad (10)$$

where α_{nf} and β_{fm} are the Lagrange multipliers for constraints $S_{nf} \geq 0$ and $T_{fm} \geq 0$, respectively. The partial derivatives of Lagrange $\mathcal{L}$ concerning S_{nf} and T_{fm} are:

$$\begin{aligned} \partial \mathcal{L} / \partial S_{nf} = & -2(DT^T)_{nf} + 2(STT^T)_{nf} \\ & + \lambda \sum_{n'=1}^N (\log \frac{S_{nf}}{S_{n'f}} + 1 - \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') + \alpha_{nf} \end{aligned} \quad (11)$$

The two sentences S_n and $S_{n'}$ are very close to each other when the $\mathbb{S}(n, n')$ is nonzero. Thus, $S_{nf}/S_{n'f}$ will be approximately 1. To reduce computational complexity, $\log(S_{nf}/S_{n'f})$ is set to 0.

$$\begin{aligned} \partial \mathcal{L} / \partial S_{nf} = & -2(DT^T)_{nf} + 2(STT^T)_{nf} \\ & + \lambda \sum_{n'=1}^N (1 - \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') + \alpha_{nf} \end{aligned} \quad (12)$$

$$\partial \mathcal{L} / \partial T_{fm} = -2(S^T D)_{fm} + 2(S^T ST)_{fm} + \beta_{fm} \quad (13)$$

Using the KKT conditions $\alpha_{nf} S_{nf} = 0$ and $\beta_{fm} T_{fm} = 0$, the following updating rules can be achieved for S_{nf} and T_{fm} :

$$S_{nf} = S_{nf} \times \frac{(DT^T)_{nf} + \lambda \sum_{n'=1}^N (S_{n'f}/S_{nf}) \mathbb{S}(n, n')}{(STT^T)_{nf} + \lambda \sum_{n'=1}^N \mathbb{S}(n, n')} \quad (14)$$

$$T_{fm} = T_{fm} \times \frac{(S^T D)_{fm}}{(S^T ST)_{fm}} \quad (15)$$

3.1 Proof of convergence

The proof follows a similar procedure in Lee and Seung's paper for the basic NMF [21]. Proof

involves a theorem that is proved using three lemmas. Respect for updating rules in Eq. (14) and Eq. (15), we can present the theorem as follows:

Theorem. The cost function of O in Eq. (9) is non-increasing regarding the updates in Eq. (14) and Eq. (15).

When $\lambda = 0$, the updating rules in Eq. (14) reduces to the basic rule of NMF. The auxiliary function like that applied in the Expectation-Maximization method is used in this paper [31].

Definition. $G(x, x^t)$ is an auxiliary function for $O(x)$ if the conditions $G(x, x^t) \geq O(x)$, and $G(x, x) = O(x)$ are satisfied.

Lemma 1. If G is an auxiliary function for O , then O is non-increasing regarding the updating rule:

$$x^{t+1} = \underset{x}{\operatorname{argmin}} G(x, x^t) \quad (16)$$

Proof. $O(x^{t+1}) \leq G(x^{t+1}, x^t) \leq G(x^t, x^t) = O(x^t)$

The equality $O(x^{t+1}) = O(x^t)$ holds if x^t is a local minimum of $G(x, x^t)$. A series of estimations can be obtained to converge to a local minimum of the cost function by repeating the updating rule in Eq. (16). Considering a suitable auxiliary function, we can demonstrate that the rule of S_{nf} in Eq. (14) is the updating rule in Eq. (16).

Lemma 2. The function

$$G_{nf}(S, S^t) = \mathcal{F}_{nf}(S^t) + \mathcal{F}'_{nf}(S^t)(S - S^t)_{nf} + \frac{(S^t T T^T)_{nf}}{S_{nf}^t} (S - S^t)_{nf}^2 + \lambda \sum_{n=1}^N \sum_{n'=1}^N \sum_{f=1}^K (S_{nf} \log \frac{S_{nf}}{S_{n'f}} + S_{n'f} \log \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') \quad (17)$$

is an auxiliary function for cost function in Eq. (9).

Proof. $G_{nf}(S, S) = O$ is clear. The following part is the difference between $O_{nf}(S)$ and $G_{nf}(S, S^t)$:

$$\sum_{n=1}^N \sum_{m=1}^M (D_{nm} - \sum_{f=1}^K S_{nf} T_{fm})^2 \quad (18)$$

The Taylor series expansion of Eq.(18) is as follows:

$$\mathcal{F}_{nf}(S^t) + \mathcal{F}'_{nf}(S^t)(S - S^t)_{nf} + (T T^T)_{ff} (S - S^t)_{nf}^2 \quad (19)$$

To find $G_{nf}(S, S^t) \geq O_{nf}(S)$, the Eq.(19) is compared with the different part of $G_{nf}(S, S^t)$:

$$\mathcal{F}_{nf}(S^t) + \mathcal{F}'_{nf}(S^t)(S - S^t)_{nf} + \frac{(S^t T T^T)_{nf}}{S_{nf}^t} (S - S^t)_{nf}^2 \quad (20)$$

To prove $G_{nf}(S, S^t) \geq O_{nf}(S)$, the following inequality can be used:

$$\frac{(S^t T T^T)_{nf}}{S_{nf}^t} \geq (T T^T)_{ff} \quad (21)$$

To demonstrate inequality (21), we have

$$(S^t T T^T)_{nf} = \sum_{k=1}^K S_{nk}^t (T T^T)_{kf} \geq S_{nf}^t (T T^T)_{ff} \quad (22)$$

Thus, inequality $G_{nf}(S, S^t) \geq O_{nf}(S)$ holds. Subsequently, an auxiliary function for the updating rule in Eq. (17) is determined.

Lemma 3. The function

$$G_{fm}(T, T^t) = O_{fm}(T^t) + O'_{fm}(T^t)(T - T^t)_{fm} + \frac{(S^T S T^t)_{fm}}{T_{fm}^t} (T - T^t)_{fm}^2 \quad (23)$$

is an auxiliary function for $O_{fm}(T)$, where O_{fm} specifies the part of O which is relevant to T_{fm} .

$$O_{fm}(T) = O_{fm}(T^t) + O'_{fm}(T^t)(T - T^t)_{fm} + (S^T S)_{ff} (T - T^t)_{fm}^2 \quad (24)$$

The proof of lemma (3) is the same as lemma (2) and is ignored for space limitation. The convergence of theorem (1) is proved by considering these lemmas.

Proof of theorem. According to $G(x, x')$ in Eq. (16) and lemma (2), we get the following updating rule:

$$\frac{\partial G_{nf}(S, S^t)}{\partial S_{nf}} = -(TT^T)_{ff}(S - S^t)_{nf} + \lambda \sum_{n'=1}^N (1 - \frac{S_{n'f}}{S_{nf}}) \mathbb{S}(n, n') = 0 \quad (25)$$

$$S_{nf}^{t+1} = S_{nf}^t \times \frac{(DT^T)_{nf} + \lambda \sum_{n'=1}^N (S_{n'f}/S_{nf}) \mathbb{S}(n, n')}{(STT^T)_{nf} + \lambda \sum_{n'=1}^N \mathbb{S}(n, n')} \quad (26)$$

Similarly, exchanging $G(x, x')$ in Eq. (16) by Eq. (23) leads to the updating rule:

$$T_{fm}^{t+1} = T_{fm}^t \times \frac{(S^T D)_{fm}}{(S^T ST)_{fm}} \quad (27)$$

Since Eq. (17) and Eq. (23) are auxiliary functions, O is non-increasing under these updating rules.

3.2 Sentence scoring

Non-negativity constraint in NMFTS provides better interpretability and enhances understanding of the document in latent semantic space than SVD [18]. Assuming a text consists of N sentences and M terms. NMFTS decomposes matrix D to two matrices S and T , where rows of D correspond to text sentences and columns show terms. Rows in T correspond to K document topics in the latent semantic space and rows in S correspond to sentences in the document. Element S_{nf} represents the strength of topic f in sentence n , and element T_{fm} signifies the contribution of term m in topic f . Lee et al. proposed NMF based method for ATS by computing Generic Relevance Score (GRS) for each sentence as follows [13]:

$$\mathbb{W}_f = \frac{\sum_{n=1}^N S_{nf}}{\sum_{n=1}^N \sum_{k=1}^K S_{nk}} \quad (28)$$

$$GRS(n) = \sum_{f=1}^K S_{nf} \mathbb{W}_f \quad (29)$$

Where $\mathbb{W}_f$ is the normalized strength of topic f in document sentences. It species the relative contribution of each topic in sentences. Top scoring

sentences are selected for creating a summary. The algorithm (1) shows the learning process of the proposed NMFTS algorithm.

Algorithm 1: The proposed NMFTS for TS

Input:

Sentence-term matrix (D)
Latent space dimension (K)
Regularizer parameter (λ)

Output:

Sentences scores

Initializing:

Initialize S and T as non-negative random matrices

for $n = 1$ **to** N **do**

for $n' = 1$ **to** N **do**

 Compute the similarity of $\mathbb{S}(n, n')$ using equation (7)

end

end

Factorizing:

while *terminal conditions not satisfied* **do**

for $n = 1$ **to** N **do**

for $f = 1$ **to** K **do**

$$S_{nf} = S_{nf} \times \frac{(DT^T)_{nf} + \lambda \sum_{n'=1}^N (S_{n'f}/S_{nf}) \mathbb{S}(n, n')}{(STT^T)_{nf} + \lambda \sum_{n'=1}^N \mathbb{S}(n, n')}$$

end

end

for $m = 1$ **to** M **do**

for $f = 1$ **to** K **do**

$$T_{fm} = T_{fm} \times \frac{(S^T D)_{fm}}{(S^T ST)_{fm}}$$

end

end

end

Scoring:

for $f = 1$ **to** K **do**

$$\mathbb{W}_f = \frac{\sum_{n=1}^N S_{nf}}{\sum_{n=1}^N \sum_{k=1}^K S_{nk}}$$

end

for $n = 1$ **to** N **do**

$$GRS(n) = \sum_{f=1}^K S_{nf} \mathbb{W}_f$$

end

return Top scoring sentences

3.3 Time complexity

The cost function of the proposed NMFTS is minimized by iteratively updating rules. Thus, time complexity is crucial for algorithm efficiency. For NMFTS, it is essential to indicate that the similarity matrix is usually sparse. We have N sentences and need $N \times N \times K$ floating-point addition and multiplication to apply constraints in update rules. Alongside the updating complexity, NMFTS also requires $O(N^2M)$ to create the similarity matrix. The similarity matrix leads to more operations to update the solution of the cost function. The time complexity of NMFTS is equal to basic NMF because the similarity matrix is sparse. Assume the update rules terminate after t iterations; the overall computations of the NMF and NMFTS are $O(NMKt)$ and $O(\max(NM, NN)Kt)$, respectively. It is worth noting that the running time depends on the number of iterations. Regarding results, the proposed NMFTS converges fast, and the overall running time is not high for the whole updating process.

4 Experiments

In this Section, a series of experiments were conducted for the performance comparison of the proposed NMFTS. Four public data sets were used in this paper: DUC2001, DUC2002, CNN, and DailyMail, which have been utilized in similar works for performance evaluation [6, 8, 11]. The following Sections explain the compared method and results.

4.1 Preprocessing

Document preprocessing is an essential phase to prepare the text for TS. In this paper, tokenization, stop words removal, and word stemming techniques are used for preprocessing. Tokenization divides the document into words or terms. Most of the words that are usually occurred in the sentences can be neglected as they are used to connect the words to make a sentence. They are assumed as stop words that do not show the content of the documents. Stemming is the process of connecting the different types of a word into a general form of the term, the stem [30].

Many document representation methods have been applied to show the texts in the processing

step. This paper uses a vector model for representation.

Bag-of-Words (BoW): This model presents each sentence in the document as a M -dimensional vector, where M indicates the number of terms in the document [32]. Each entry value in the BoW vector shows the number of occurrences of the term in the sentence multiplied by the IDF of the term. This model has a high dimensionality, and it does not contain word dependency. Since most input values are zeros in the vector, it is sparse. It neglects syntax and ignores the meaning of words. Semantic representation of documents is inaccurate because the arrangement and order of terms are neglected.

Vector space model (VSM) or term vector: this model represents sentences as vectors [33]. VSM is typically employed in information retrieval (IR) by modeling texts as weighted vectors. The text contains a set of M words. Each word has its corresponding weight, and each weight shows the importance of the corresponding word in the text. The weights are calculated using the frequency of the words in the text, term frequency-inverse document frequency (TF-IDF) [34]. The advantage of VSM is the simplicity of searching for sentences or comparing sentences using one of the similarity or distance criteria. The BoW model is the popular VSM method due to its simplicity and general application.

Word embedding: NNs are applied to model terms as vectors. Various techniques are used to learn the word embedding like the Word2Vector model [35]. In this model, each word based on the meaning of the word is embedded as a real-valued vector in the Euclidean space. Recently, new kinds of representation have been introduced such as sentence, paragraph, or text embeddings.

4.2 Performance evaluation measure

The performance of the proposed approach is measured using a recall-oriented understudy for gisting evaluation (ROUGE) toolkit [36]. ROUGE measures recall, precision, and F-measure of candidate summaries against reference summaries. While recall evaluates how much of the reference summary is captured by created summary, precision measures how much of the generated summary is appropriate. However, the limitation in the se-

lection of created summary length makes precision less significant. Recall computes the ratio between the number of co-occurring n -grams seen in both the candidate and reference summaries and the total number of n -grams in the reference summary. ROUGE is calculated as follows:

$$ROUGE = \frac{\sum_{gram_n \in \{reference\ summary\}} Count_{match}(gram_n)}{\sum_{gram_n \in \{reference\ summary\}} Count(gram_n)} \quad (30)$$

where n is the length of the n -gram and $Count_{match}(gram_n)$ is the maximum number of n -grams overlapping in both created and reference summaries [36]. This paper used recall scores of ROUGE-1, ROUGE-2, and ROUGE-L to evaluate the performance of the proposed model. ROUGE-1 computes co-occurring uni-grams between candidate and reference summaries, ROUGE-2 estimates bi-grams, while ROUGE-L computes the proportion between the summaries' longest common subsequence (LCS) length and the reference summary length. Recall scores for these three ROUGE variations are computed for DUC2001, DUC2002 documents while F-measure is reported for CNN and DailyMail documents.

4.3 Results and discussions

This paper performed experiments on the benchmark data sets to assess the effectiveness of the proposed method compared with the other techniques. All experiments are performed on a machine with an Intel Core i7 processor, 16G RAM, and Windows 7 (64-bit).

The proposed NMFTS has two parameters: the latent space dimension and the regularization parameter (λ). Finding optimal value for latent space dimension (K) is impossible. This paper follows the approach presented by Aliguliyev et al.[37] to specify the latent space dimension in a text. This method is based on the distribution of terms in the sentences. K for non-negative matrix factorization of sentence-term matrix D is computed as:

$$K = \frac{N \times M}{\#terms\ in\ D} \quad (31)$$

In this paper, the trial and error method is used to find the optimal value for λ .

4.3.1 DUC2001 and DUC2002 datasets

DUC2001 corpus consists of 308 texts while the DUC2002 dataset contains 567 texts for generic single text summarization. Reference summaries for both datasets are abstractive and are about 100 terms. Each text in both datasets has 1–3 reference summaries.

Table (1) shows the detailed summarization results on the DUC2002 corpus. NMFTS demonstrates extraordinary performance. It reached ROUGE-1=53.1, ROUGE-2=26.2, and ROUGE-L=46.4, which demonstrates that similarity regularizers between sentences help improve the summarization performance.

The basic NMF_F and NMF_{KL} fail to enhance the ROUGE after the 100th repetition and perform worse than many compared methods. The original NMF methods cannot keep similarities between sentences in the latent semantic space.

The basic NMF_F and NMF_{KL} obtain competitive ROUGE-1 and ROUGE-2 scores compared to the URANK approach [38]. As the ROUGE-L score for the URANK technique has not been reported, its superiority is uncertain. Also, URANK follows an integrated method for summarizing single and multiple texts at the same time, making it inappropriate for a single TS stand-alone task.

TGRAPH conducts topic modeling utilizing external knowledge and converts the text to a weighted graph [39]. They optimize coherence, importance, and redundancy for TS by applying integer linear programming. TGRAPH approach depends on domain and collection as it utilizes external knowledge to create topics for the graph generation. Thus, this approach is inappropriate for generic TS.

CoRank and CoRank+ apply relationships between sentences and between words using the graph-based method for sentence scoring [8]. Both approaches are unsupervised and independent language, collection, and domain similar to the proposed NMFTS. As seen in Table (1) CoRank+ outperforms compared methods, however, NMFTS is superior to CoRank+ method.

COSUM performs summarization as a clustering task [6]. ROUGE-1 and ROUGE-2 performance of COSUM outperform basic NMF meth-

Table 1. The performance of the compared methods on the DUC2002 dataset.

	ROUGE-1	ROUGE-2	ROUGE-L
NMF _F	46.3	18.2	41.5
NMF _{KL}	47.6	19.7	42.4
URANK	48.5	21.5	-
TGRAPH	48.1	24.3	-
CoRank	50.7	24.0	43.4
CoRank+	52.6	25.8	45.1
COSUM	49.1	23.1	-
iGraph	48.5	22.0	43.7
iGraph-R	49.2	23.1	44.1
NN-SE	47.4	23.0	43.5
Deep-Classifier	46.8	22.6	43.1
SummaRuNNer	46.6	23.1	43.0
HSSAS	52.1	24.5	48.8
DQN	46.4	22.7	42.9
NMFTS	53.1	26.2	46.4

ods. However, in the absence of the ROUGE-L metric and assessment on other data sets, the overall score of COSUM is not decisively better. Also, COSUM employs stemming and therefore is language-dependent.

graph-based approaches, iGraph and iGraph-R, employ an embedding method to find the latent semantic properties at the word, bigram, and trigram levels [5]. The embedding methods are trained using extracted words with part-of-speech tags, bigrams, and trigrams. The embedding method is utilized to compute the similarity between sentences that act as graph links and then TextRank is employed to rank sentences [40]. The proposed NMFTS performance is better than the iGraph performance. Both iGraph and iGraph-R apply external knowledge to train models to generate similarities between sentences which prevents use for generic TS.

Table 2. The performance of the compared methods on the DUC2001 dataset.

	ROUGE-1	ROUGE-2	ROUGE-L
NMF _F	42.1	14.2	37.4
NMF _{KL}	44.7	15.6	39.3
URANK	45.4	17.6	-
COSUM	47.3	20.1	-
NMFTS	50.4	23.3	43.2

Results demonstrate that NMFTS outperforms HSSAS [41], NN-SE [42], Deep-Classifier [43],

SummaRuNNer [11] and DQN [44] for ROUGE-1, ROUGE-2, and ROUGE-L. Unlike the proposed NMFTS, these techniques are supervised methods [11]. So, their suitability for generic TS is arguable.

Table (2) displays the detailed summarization results on the DUC2001 data set. As seen in this table, the proposed NMFTS method outperforms all the other approaches in terms of ROUGE-1, ROUGE-2, and ROUGE-L, which confirms NMFTS can learn the latent semantic of texts.

Due to the unavailability of ROUGE-L scores for both the URANK and COSUM techniques, there is no clear winner. TGRAPH, Corank, and iGraph are not evaluated in this dataset. Surprisingly, none of the supervised approaches reported in the Table (1) in the DUC2001 dataset were evaluated.

4.3.2 CNN and DailyMail data sets

The documents of CNN and DailyMail datasets are divided into training, validation, and test sets. CNN dataset includes 92,579 texts whereas the DailyMail dataset comprises 219,506 texts. The proposed NMFTS approach is evaluated using a test set for both corpora taken together. Each text in CNN and DailyMail corpus has story highlights that are utilized as reference summaries. For comparing with state-of-the-art approaches that used CNN and DailyMail data sets [45, 46], three sentences for each text in CNN and four sentences for each text in DailyMail are extracted as summaries.

Table 3. The performance of the compared methods on the CNN and DailyMail datasets.

	ROUGE-1	ROUGE-2	ROUGE-L
NMF _F	30.8	10.4	27.7
NMF _{KL}	30.4	10.9	27.4
NN-SE	35.5	14.7	32.2
NEUSUM	41.6	19.0	38.0
SummaRuNNer	39.6	16.2	35.3
REFRESH	40.0	18.2	36.6
HSSAS	42.3	17.8	37.6
DQN	39.4	16.1	35.6
BANDITSUM	41.5	18.7	37.6
NMFTS	42.6	18.8	37.9

Table (3) shows comparison results of the proposed NMFTS and compared approaches for combined CNN and DailyMail datasets. The quality of NMFTS summaries is better than the NMF_F and NMF_{KL}. None of the unsupervised techniques listed in Table (1) were evaluated in this dataset.

In this paper, NN-SE, NEUSUM [46], SummaRuNNer, REFRESH [46], HSSAS, DQN, and BANDITSUM [47] techniques are applied as compared methods to combined CNN and DailyMail corpus. All these approaches use a variation of the deep learning method. The hyper-parameters of these supervised approaches are tuned using the validation set.

Table (3) displays that all compared approaches create acceptable results on the CNN and DailyMail data set. Though effectiveness of deep learning methods for extractive TS is indicated, the performance of the trained models on other domains requires further investigation. For example, the trained SummaRuNNer model on CNN and DailyMail data set achieved a better score than basic NMF in Table (3), but the same model demonstrates relatively lower performance on out-of-domain DUC2002 corpus.

According to this table, NMFTS considerably outperforms the other summarization methods regarding ROUGE-1. The ROUGE-2 and ROUGE-L values of the proposed method were slightly lower than the NEUSUM method. It indicates the importance of constraints in learning the semantics of texts.

5 Conclusions

This paper proposed a novel non-negative matrix factorization for text summarization (NMFTS). The proposed method used the pairwise similarity between sentences to enhance the performance of TS. The similarity regulators are imposed on the objective function to improve the convergence speed. The complete proof of convergence and experimental results on the benchmark data sets have confirmed the effectiveness of the proposed approach. NMFTS could learn a better representation by leveraging the power of both the parts-based representation and constraints. The proposed NMFTS can outperform other methods by setting the regularization parameter λ . As future work, the plan is to evaluate the proposed method in topic modeling. Another goal is to investigate the possibility of a generalized version of the proposed approach. The selection of optimum value for the regularized parameter is a challenge for the proposed method, and the plan is to develop a strategy for the automatic selection of λ .

Acknowledgments

The author would like to thank the University of Bonab for financial and technical support.

References

- [1] M. H. Aghdam, S. Heidari, Feature selection using particle swarm optimization in text categorization, *Journal of Artificial Intelligence and Soft Computing Research* 5 (4) (2015) 231–238.
- [2] G. C. V. Vilca, M. A. S. Cabezudo, A study of abstractive summarization using semantic representa-

- tions and discourse level information (2017) 482–490.
- [3] N. Moratanch, S. Chitrakala, A survey on extractive text summarization, in: 2017 international conference on computer, communication and signal processing (ICCCSP), IEEE, 2017, pp. 1–6.
 - [4] M. Gambhir, V. Gupta, Recent automatic text summarization techniques: a survey, *Artificial Intelligence Review* 47 (1) (2017) 1–66.
 - [5] K. Yang, K. Al-Sabahi, Y. Xiang, Z. Zhang, An integrated graph model for document summarization, *Information* 9 (9) (2018) 232.
 - [6] R. M. Alguliyev, R. M. Aliguliyev, N. R. Isazade, A. Abdi, N. Idris, Cosum: Text summarization based on clustering and optimization, *Expert Systems* 36 (1) (2019) e12340.
 - [7] E. Lloret, M. T. Romá-Ferri, M. Palomar, Compendium: A text summarization system for generating abstracts of research papers, *Data & Knowledge Engineering* 88 (2013) 164–175.
 - [8] C. Fang, D. Mu, Z. Deng, Z. Wu, Word-sentence co-ranking for automatic extractive text summarization, *Expert Systems with Applications* 72 (2017) 189–195.
 - [9] M. A. Fattah, F. Ren, Ga, mr, ffnn, pnn and gmm based models for automatic text summarization, *Computer Speech & Language* 23 (1) (2009) 126–144.
 - [10] D. Shen, J.-T. Sun, H. Li, Q. Yang, Z. Chen, Document summarization using conditional random fields., in: *IJCAI*, Vol. 7, 2007, pp. 2862–2867.
 - [11] R. Nallapati, F. Zhai, B. Zhou, Summarunner: A recurrent neural network based sequence model for extractive summarization of documents, in: *Thirty-first AAAI conference on artificial intelligence*, 2017.
 - [12] O. Vikas, A. K. Meshram, G. Meena, A. Gupta, Multiple document summarization using principal component analysis incorporating semantic vector space model (2008) 141–156.
 - [13] J.-H. Lee, S. Park, C.-M. Ahn, D. Kim, Automatic generic document summarization based on non-negative matrix factorization, Vol. 45, Elsevier, 2009, pp. 20–34.
 - [14] W. S. El-Kassas, C. R. Salama, A. A. Rafea, H. K. Mohamed, Automatic text summarization: A comprehensive survey, *Expert Systems with Applications* 165 (2021) 113679.
 - [15] D. Sahoo, R. Balabantaray, M. Phukon, S. Saikia, Aspect based multi-document summarization, in: *2016 International Conference on Computing, Communication and Automation (ICCCA)*, IEEE, 2016, pp. 873–877.
 - [16] M. J. Mohan, C. Sunitha, A. Ganesh, A. Jaya, A study on ontology based abstractive summarization, *Procedia Computer Science* 87 (2016) 32–37.
 - [17] M. Mohd, R. Jan, M. Shah, Text document summarization using word embedding, *Expert Systems with Applications* 143 (2020) 112958.
 - [18] D. D. Lee, H. S. Seung, Learning the parts of objects by non-negative matrix factorization, *Nature* 401 (6755) (1999) 788.
 - [19] M. H. Aghdam, M. D. Zanjani, A novel regularized asymmetric non-negative matrix factorization for text clustering, *Information Processing & Management* 58 (6) (2021) 102694.
 - [20] M. H. Aghdam, A novel constrained non-negative matrix factorization method based on users and items pairwise relationship for recommender systems, *Expert Systems with Applications* (2022) 116593.
 - [21] D. D. Lee, H. S. Seung, Algorithms for non-negative matrix factorization, in: *Advances in neural information processing systems*, 2001, pp. 556–562.
 - [22] D. Cai, X. He, J. Han, T. S. Huang, Graph regularized nonnegative matrix factorization for data representation, *IEEE transactions on pattern analysis and machine intelligence* 33 (8) (2010) 1548–1560.
 - [23] H. Liu, Z. Wu, X. Li, D. Cai, T. S. Huang, Constrained nonnegative matrix factorization for image representation, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 34 (7) (2011) 1299–1311.
 - [24] X. Luo, M. Zhou, Y. Xia, Q. Zhu, An efficient non-negative matrix-factorization-based approach to collaborative filtering for recommender systems, *IEEE Transactions on Industrial Informatics* 10 (2) (2014) 1273–1284.
 - [25] X. Luo, M. Zhou, S. Li, Z. You, Y. Xia, Q. Zhu, A nonnegative latent factor model for large-scale sparse matrices in recommender systems via alternating direction method, *IEEE transactions on neural networks and learning systems* 27 (3) (2015) 579–592.
 - [26] O. Vikas, A. K. Meshram, G. Meena, A. Gupta, Multiple document summarization using principal component analysis incorporating semantic vector space model, in: *International Journal of Computational Linguistics & Chinese Language Processing*,

- Volume 13, Number 2, June 2008, 2008, pp. 141–156.
- [27] Y. Gong, X. Liu, Generic text summarization using relevance measure and latent semantic analysis, in: Proceedings of the 24th annual international ACM SIGIR conference on Research and development in information retrieval, 2001, pp. 19–25.
- [28] J. Qiang, Y. Li, Y. Yuan, W. Liu, Snapshot ensembles of non-negative matrix factorization for stability of topic modeling, *Applied Intelligence* 48 (11) (2018) 3963–3975.
- [29] C. Liu, Discriminant analysis and similarity measure, *Pattern Recognition* 47 (1) (2014) 359–367.
- [30] C. C. Aggarwal, C. Zhai, Mining text data, Springer Science & Business Media, 2012.
- [31] A. P. Dempster, N. M. Laird, D. B. Rubin, Maximum likelihood from incomplete data via the em algorithm, *Journal of the Royal Statistical Society: Series B (Methodological)* 39 (1) (1977) 1–22.
- [32] G. Erkan, D. R. Radev, Lexrank: Graph-based lexical centrality as salience in text summarization, *Journal of artificial intelligence research* 22 (2004) 457–479.
- [33] A. Ibrahim, T. Elghazaly, M. Gheith, A novel arabic text summarization model based on rhetorical structure theory and vector space model, *International Journal of Computational Linguistics and Natural Language Processing* 2 (8) (2013) 480–485.
- [34] G. Salton, C. Buckley, Term-weighting approaches in automatic text retrieval, Vol. 24, Elsevier, 1988, pp. 513–523.
- [35] O. Mogren, M. Kågebäck, D. Dubhashi, Extractive summarization by aggregating multiple similarities, in: Proceedings of the International Conference Recent Advances in Natural Language Processing, 2015, pp. 451–457.
- [36] C.-Y. Lin, Rouge: A package for automatic evaluation of summaries, in: Text summarization branches out, 2004, pp. 74–81.
- [37] R. M. Aliguliyev, A new sentence similarity measure and sentence based extractive technique for automatic text summarization, *Expert Systems with Applications* 36 (4) (2009) 7764–7772.
- [38] X. Wan, Towards a unified approach to simultaneous single-document and multi-document summarizations, in: Proceedings of the 23rd international conference on computational linguistics (Coling 2010), 2010, pp. 1137–1145.
- [39] D. Parveen, H.-M. Ramsel, M. Strube, Topical coherence for graph-based extractive summarization, in: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing, 2015, pp. 1949–1954.
- [40] R. Mihalcea, P. Tarau, Textrank: Bringing order into text, in: Proceedings of the 2004 conference on empirical methods in natural language processing, 2004, pp. 404–411.
- [41] K. Al-Sabahi, Z. Zuping, M. Nadher, A hierarchical structured self-attentive model for extractive document summarization (hssas), *IEEE Access* 6 (2018) 24205–24212.
- [42] J. Cheng, M. Lapata, Neural summarization by extracting sentences and words, *arXiv preprint arXiv:1603.07252* (2016).
- [43] R. Nallapati, B. Zhou, M. Ma, Classify or select: Neural architectures for extractive document summarization, *arXiv preprint arXiv:1611.04244* (2016).
- [44] K. Yao, L. Zhang, T. Luo, Y. Wu, Deep reinforcement learning for extractive document summarization, *Neurocomputing* 284 (2018) 52–62.
- [45] S. Narayan, S. B. Cohen, M. Lapata, Ranking sentences for extractive summarization with reinforcement learning, *arXiv preprint arXiv:1802.08636* (2018).
- [46] Q. Zhou, N. Yang, F. Wei, S. Huang, M. Zhou, T. Zhao, Neural document summarization by jointly learning to score and select sentences, *arXiv preprint arXiv:1807.02305* (2018).
- [47] Y. Dong, Y. Shen, E. Crawford, H. van Hoof, J. C. K. Cheung, Banditsum: Extractive summarization as a contextual bandit, *arXiv preprint arXiv:1809.09672* (2018).



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